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Chapter 19

Router Essentials and Route Summarization

In this chapter, you are presented with questions that test you on essential routing concepts in order to prepare you for the ICND2 exam. This chapter reviews basic information regarding routing protocols and also tests your knowledge of calculating summary routes, given detail route information.

The Problems You'll Work On

In this chapter, you'll review questions concerning the following topics:

- Understanding static and dynamic routing
- Recognizing routing terminology
- Working with route summarization
- Knowing about inter-VLAN routing

What to Watch Out For

Don't let common mistakes trip you up; watch for the following when working with these questions:

- For the ICND2 exam, be sure to know the benefits and drawbacks of both static routing and dynamic routing.
- Know the basics of routing protocols such as distance vector versus link state routing protocols.
- Know the administrative distances associated with each of the routing protocols.
- Know that route summarization is designed to keep a small routing table and how to calculate a summary route, given the actual routes.
- You will be tested heavily on routing between VLANs and how to configure a router on a stick. Know that on the router, you need to configure sub-interfaces and set the encapsulation protocol for each sub-interface.

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Understanding Static and Dynamic Routing

776–797 Choose the best answer(s).

776. Which of the following are true of RIPv1? (Select two.)

(A) Administrative distance of 90

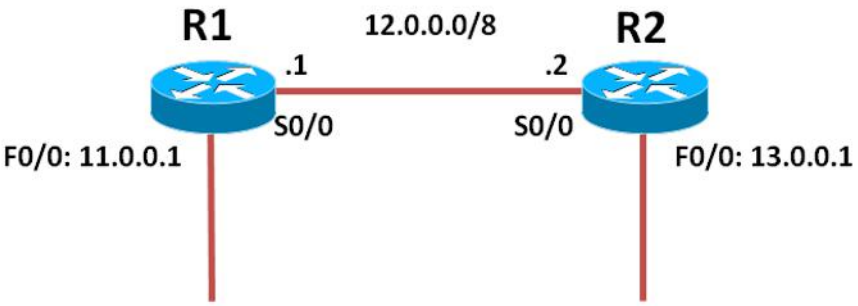
(B) Is classful

(C) Administrative distance of 120

(D) Is classless

(E) Vendor specific

777. You have configured a network as shown in the figure below. From router R1, you can ping the S0/0 interface on router R2, but you cannot ping the F0/0 interface on router R2. What is the cause of the problem?



(A) The default gateway of router R2 is incorrect.

(B) There is no route for the 13.0.0.0 network on router R1.

(C) You need to load RIP on router R2.

(D) The default gateway on router R1 is incorrect.

778. Which of the following is true of classful routing protocols?

(A) They send subnet mask info with the routing update.

(B) They use wildcard masks.

(C) They use the route with the highest cost pathway.

(D) They do not send the subnet mask info with the routing update.

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779. Which of the following is true of RIPv2? (Select two.)

(A) Administrative distance of 90

(B) Is classful

(C) Administrative distance of 120

(D) Is classless

(E) Vendor specific

780. Distance vector routing protocols use which of the following as their metric?

(A) Bandwidth

(B) Link speed

(C) Hop count

(D) Available bandwidth

781. What is the administrative distance of RIP?

(A) 90

(B) 100

(C) 110

(D) 120

(E) 1

782. Which of the following is true of EIGRP? (Select two.)

(A) Uses cost as its metric

(B) Is vendor-specific

(C) Is an open standard

(D) Administrative distance of 110

(E) Administrative distance of 90

783. Which of the following identifies the benefit of static routing over dynamic routing? (Select two.)

(A) More secure because the administrator must add the routes.

(B) Less secure because the administrator must add the routes.

(C) Bandwidth is being used to update the routing tables.

(D) No bandwidth is being used to update the routing tables.

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784. What is the default administrative distance of a static route?

(A) 90

(B) 100

(C) 110

(D) 120

(E) 1

785. Which of the following is true of OSPF? (Select two.)

(A) Elects a DR

(B) Is vendor-specific

(C) Administrative distance of 90

(D) Uses cost as its metric

(E) Administrative distance of 110

786. What is the administrative distance of OSPF?

(A) 90

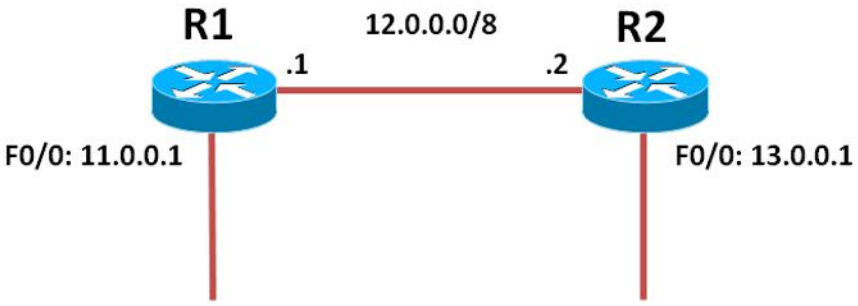
(B) 100

(C) 110

(D) 120

(E) 1

787. You are the network administrator for a small network that has two routers (R1 and R2, as shown in the figure below) that are connected via serial interfaces. You have used the command `Ip route 13.0.0.0 255.0.0.0 12.0.0.2` on R1 to finalize configuration. What happens if the serial interface on R2 shuts down?



(A) A triggered update is sent from R2 to R1.

(B) A triggered update is sent from R1 to R2.

(C) The route of 13.0.0.0 remains on router R1.

(D) A poison reverse packet is sent from R2 to R1.

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788. You have no static routes configured on the router. A router learns about a route to the same network from OSPF, EIGRP, and RIP. Which route is added to the routing table?

(A) The RIP route

(B) The EIGRP route

(C) The OSPF route

(D) All the routes

789. Which of the following is a benefit of triggered updates?

(A) It decreases convergence time by sending out an update as soon as there is a change to the network topology.

(B) They are more accurate than scheduled updates.

(C) They are supported by all routing protocols.

(D) It increases convergence time by sending out an update as soon as there is a change to the network topology.

790. You are troubleshooting communication on router R1. You use the show ip route command to display the routing table. Using the output of the figure below, how will information be sent from R1 to a system with the IP address of 15.0.0.25?

```
R1(config)#do show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    11.0.0.0/8 is directly connected, FastEthernet0/0
C    12.0.0.0/8 is directly connected, Serial0/3/0
D    13.0.0.0/8 [90/2172416] via 12.0.0.2, 00:00:56, Serial0/3/0
C    14.0.0.0/8 is directly connected, Serial0/3/1
D    15.0.0.0/8 [90/2681856] via 12.0.0.2, 00:00:56, Serial0/3/0
                [90/2681856] via 14.0.0.2, 00:00:45, Serial0/3/1
```

(A) Directly through FastEthernet 0/0

(B) Always through 13.0.0.2

(C) Always through 12.0.0.2

(D) Through 12.0.0.2, and through 14.0.0.2

(E) Always through 14.0.0.2

791. Which of the following is true of link state routing protocols? (Select two.)

(A) Choose the route with the lowest hop count

(B) They use more resources as they store multiple tables in memory

(C) Use a single table to share routing information

(D) Use triggered updates to reduce the time to converge

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792. You would like to ensure that a static route is used as a backup over a route using the same pathway discovered by a dynamic routing protocol. What parameter would you change?

(A) Cost

(B) Bandwidth

(C) Administrative distance

(D) Clock rate

793. What are the three tables that a link state routing protocol typically uses? (Select three.)

(A) Topology table

(B) MAC table

(C) Neighbor table

(D) STP table

(E) Routing table

794. Which of the following routing protocols does not auto summarize routes?

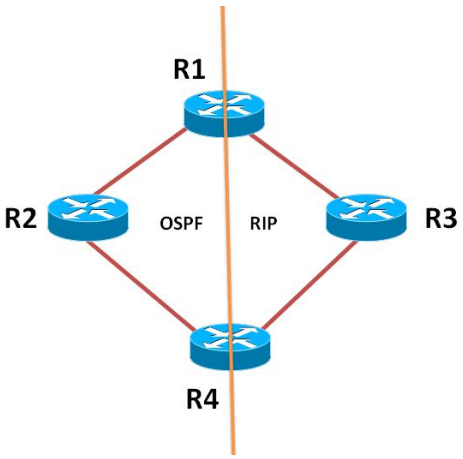
(A) RIP

(B) OSPF

(C) EIGRP

(D) IGRP

795. Looking at the figure below, when router R1 sends data to router R4, what pathway will be used?



(A) R1 cannot reach R4.

(B) Through R3.

(C) Through R2.

(D) R1 will drop all packets for R4.

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796. Which of the following commands alters the administrative distance on the static route from the default value?

(A) `ip route 21.0.0.0 255.0.0.0 20.0.0.2`

(B) `ip route 21.0.0.0 255.0.0.0 20.0.0.2 admin-distance 110`

(C) `admin-distance static route 21.0.0.0 255.0.0.0 20.0.0.2 110`

(D) `ip route 21.0.0.0 255.0.0.0 20.0.0.2 110`

797. You would like to enable IPv6 routing on your Cisco router. What command would you use?

(A) `ip route ipv6`

(B) `ip route local ipv6`

(C) `ipv6 forwarding`

(D) `ipv6 unicast-routing`

Recognizing Routing Terminology

798–803 Choose the best answer(s).

798. Which of the following does a link state routing protocol use to build the topology database? (Select two.)

(A) Hop count

(B) Hello messages

(C) Static routes

(D) BPDU from other devices

(E) LSA from other routers

799. Which of the following is the term for packets that are flooded when a topology change occurs?

(A) Holddown timer

(B) Poison reverse

(C) Count to infinity

(D) LSA

(E) Split horizon

800. Which of the following causes a router to ignore updates from lower metric paths for a period of time?

(A) Holddown timer

(B) Poison reverse

(C) Count to infinity

(D) LSA

(E) Split horizon

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801. Which of the following is the term for preventing a router from sending updates on a route through the interface on which it received the knowledge of the route?

(A) Holddown timer

(B) Poison reverse

(C) Count to infinity

(D) LSA

(E) Split horizon

802. Which of the following involves a router learning that a route is down from its neighbor, and then sending an update back to the neighbor on the route with an infinite metric?

(A) Holddown timer

(B) Poison reverse

(C) Count to infinity

(D) LSA

(E) Split horizon

803. Which of the following routing protocols are vendor specific?

(A) RIP

(B) EIGRP

(C) OSPF

(D) ARP

Working with Route Summarization

804–808 Choose the best answer(s).

804. What is the goal of route summarization?

(A) To support different size subnets

(B) To share routing table information other routers

(C) To prevent loops on the network

(D) To reduce the size of the routing table

805. You are configuring OSPF and need to determine the summary route for networks 131.107.16.0 up to 131.107.19.0. Which of the following identifies the summary route?

(A) 131.107.0.0/22

(B) 131.107.16.0/24

(C) 131.107.16.0/21

(D) 131.107.16.0/22

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806. You are configuring OSPF and need to determine the summary route for networks 172.16.0.0 up to 172.16.7.0. Which of the following identifies the summary route?

(A) 172.16.0.0/20

(B) 172.16.0.0/21

(C) 172.16.0.0/22

(D) 172.16.0.0/23

807. Which of the following identifies the summary route for networks 10.4.0.0 up to 10.7.0.0?

(A) 10.0.0.0/14

(B) 10.0.0.0/15

(C) 10.4.0.0/14

(D) 10.4.0.0/13

808. Which of the following identifies the summary route for networks 120.12.0.0 to 12.15.0.0?

(A) 120.12.0.0/16

(B) 120.12.0.0/14

(C) 120.12.0.0/23

(D) 120.12.0.0/13

Knowing about Inter-VLAN Routing

809–813 Choose the best answer(s).

809. You have a router connected to a switch that has three VLANs. You would like to configure the router so that it can be used to route traffic between the three VLANs. What do you need to do?

(A) Add two more routers to the network — one for each VLAN

(B) Enable RIP

(C) Configure router on a stick

(D) Configure OSPF

810. Which of the following switch commands would you use on the port that is connected to a router on a stick?

(A) SW1(config-if)#`switchport mode trunk`

(B) SW1(config-if)#`switchport mode access`

(C) SW1(config-if)#`switchport mode sub-interface`

(D) SW1(config-if)#`switchport port-security`

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811. Router R1 is connected to a switch that has three VLANs. You are configuring the router to route traffic between the three VLANs. What do you need to do? (Select two.)

(A) Configure the interface that is connected to the switch as a trunk port using the same encapsulation protocol.

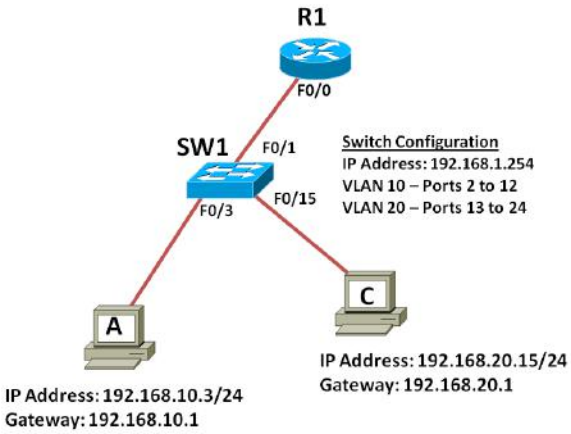
(B) Enable RIP on the router.

(C) Configure the interface that is connected to the switch as an access port.

(D) Configure the interface that is connected to the switch with three sub-interfaces.

(E) Use different encapsulation protocols on the switch and router.

812. Looking at the figure below, how would you configure the router and switch to route between the VLANs? (Select two.)



(A) On R1 perform the following commands:

R1(config)#`interface f0/0`

R1(config-if)#`no shutdown`

R1(config)#`interface f0/0.10`

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```
R1(config-subif)#encapsulation trunk 10
R1(config-subif)#ip address 192.168.10.1 255.255.255.0
R1(config)#interface f0/0.20
R1(config-subif)#encapsulation trunk 20
R1(config-subif)#ip address 192.168.20.1 255.255.255.0
```

(B) On R1 perform the following commands:

```
R1(config)#interface f0/0
R1(config-if)#no shutdown
R1(config)#interface f0/0.10
R1(config-subif)#encapsulation dot1q 10
R1(config-subif)#ip address 192.168.10.1 255.255.255.0
R1(config)#interface f0/0.20
R1(config-subif)#encapsulation dot1q 20
R1(config-subif)#ip address 192.168.20.1 255.255.255.0
```

(C) On SW1 perform the following commands:

```
SW1(config)#interface f0/1
SW1(config-if)#switchport mode access
```

(D) On SW1 perform the following commands:

```
SW1(config)#interface f0/1
SW1(config-if)#switchport mode trunk
```

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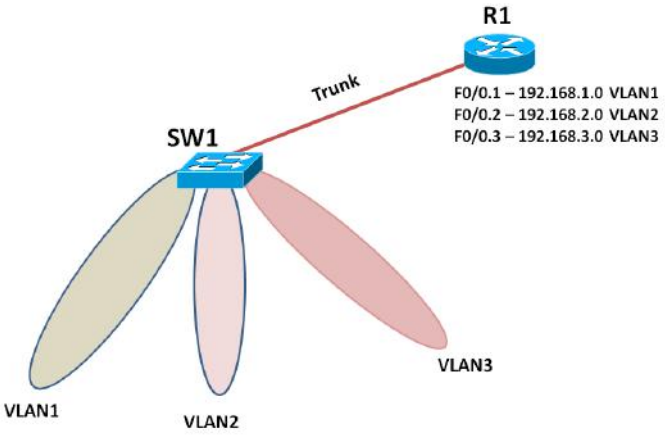
813. Sue has been assigned the task of creating a router-on-a-stick network. She has configured what is shown in the figure below. Assuming all interfaces have been properly configured, what else needs to be done?



(A) Nothing

(B) Perform the following commands on the router:

```
Router rip
Network 192.168.1.0
Network 192.168.2.0
Network 192.168.3.0
```

(C) Change the trunk port to an access port

(D) Perform the following commands on the router:

```
router eigrp 1
Network 192.168.1.0
Network 192.168.2.0
Network 192.168.3.0
```